

Nathan James Davis

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EDUCATION

University of Central Florida, Orlando, FL · B.S. in Computer Science, GPA 3.4 Expected Dec. 2027

Coursework: Data Structures & Algorithms, Systems Software, Machine Learning, Object-Oriented Programming, Discrete Structures, Linear Algebra

Honors: UCF Scholars Award, Dean's List · **Involvement:** Knight Hacks

EXPERIENCE

Repair Technician, United Data Technologies · Orlando, FL Mar. 2026 – Present

- Built a React and TypeScript browser extension that automates Odoo repair workflows: one-click ticket dating and technician assignment, device-specific canned logging responses, and batch parts-recycling updates
- Cut 2–3 minutes of manual data entry per repair and raised per-technician throughput ~10% over a 13-repair daily baseline
- Diagnose and repair hardware and software faults across a wide range of laptop makes and models, including component-level work on displays, batteries, keyboards, storage drives, and system boards

Founder & Lead Engineer, Cup of Malice Studio (independent game studio) · Orlando, FL Jul. 2026 – Present

- Lead a five-person team (two artists, a woodworker, a marketer); own all software and hardware engineering and delegate art, fabrication, and media production
- Set technical direction and manage studio infrastructure and operating costs; raised \$500+ in crowdfunding over a single weekend while exhibiting at Open Sauce 2026

PROJECTS

UVULITES — NFC Tabletop Card Game · C, Raylib, SQLite, Arduino, I2C Feb. 2026 – Present

1st Place (Judges' Vote) and **2nd Place (Crowd Favorite)** of 35 teams, UCF Project Launch 2026

Open Sauce 2026 Exhibitor · **Provisional patent filed**

- Built a two-player strategy game in C and Raylib in which players scan physical NFC cards to summon units and trigger spell effects; sustains 60 FPS on a Raspberry Pi 5 (4 GB, integrated graphics) while simulating 100+ autonomous combat entities
- Engineered a six-reader NFC array (two Arduino Nanos, each driving three PN532 modules through a TCA9548A I2C multiplexer) that reports to the host over a custom binary serial protocol with XOR checksums and a hardware watchdog
- Cut worst-case scan-to-response latency by ~80 ms through I2C clock tuning, removal of redundant passive-activation retries, and IRQ-driven asynchronous reads
- Ran continuously for 16 hours across two days at Open Sauce 2026, serving 300+ players with no crashes or manual resets
- Gate every feature on an automated test suite covering game logic and the binary serial protocol layer
- Built split-screen rendering with rotated player views, procedural biome maps, layered card sprites, and sprite animation, backed by an SQLite UID-to-card layer and internal tooling for card editing, enrollment, and scan debouncing

uvulites.com — Storefront & Membership Platform · Next.js, TypeScript, React, Shopify, Neon Postgres, Railway 2026

- Designed and built the game site and storefront in Next.js (App Router) and TypeScript, with a custom catalog and cart that hand off to Shopify checkout and a premium subscription tier
- Engineered a membership backend split across two services, syncing subscription state over HMAC-signed bidirectional channels between a Next.js entitlement engine, Neon Postgres, and a custom Shopify app
- Automated purchase, cancellation, and refund lifecycles with webhook queuing and nightly reconciliation, propagating changes instantly to Discord roles, Shopify tags, and marketing segments
- Cut billed database compute 95% (24 h/day to a measured 1.1 h/day) by removing continuous polling so the usage-billed Neon instance autosuspends when no orders or role updates are pending

Compiler Back-End · C++, ANTLR, x86-64 Assembly 2026

- Wrote a back end that emits x86-64 assembly from an ANTLR-parsed intermediate representation: stack-frame layout, the System V calling convention, and arithmetic and control-flow lowering

Social Media Follower Display · Odin, Python, Meta Graph API 2026

- Built a live follower counter for a 1920×515 ultrawide display with an animated Odin front end that reacts to count changes; deployed at an Open Sauce 2026 exhibition booth
- Wrote a Python service polling the Meta Graph API every 60 s with rate-limit budgeting, async retries, and stale-data detection; a JSON handoff makes the front end platform-agnostic and keeps it serving last-known-good data through outages

TECHNICAL SKILLS

Languages C, C++, TypeScript, Python, Java, JavaScript, SQL, Odin, x86-64 Assembly

Web & Frameworks React, Next.js, Node.js, REST APIs, Tailwind CSS, Raylib, HTML, CSS

Tools & Databases Git, Docker, Linux, CI/CD, automated testing, Neovim, JetBrains IDEs, LaTeX, PostgreSQL, SQLite

Embedded & Design Arduino, PN532 NFC, I2C, UART, serial protocol design, Onshape, 3D printing, Aseprite